

Module 11: Genomics & Biotechnology — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Explain how scientists cut, copy, and sort DNA
2. Describe how DNA fingerprinting identifies individuals
3. Explain what a GMO is and give an example
4. Describe what CRISPR does and why it is significant
5. Connect biotechnology tools back to the central dogma (Module 07)

The Big Picture

Biotechnology takes what we know about DNA and puts it to work. Because the genetic code is **universal** (the same codons mean the same amino acids in bacteria, plants, and humans — remember the central dogma from Module 07), we can move genes between organisms and they still work. That single fact makes all of genetic engineering possible.

Key Terms

The DNA Toolkit — Cut, Copy, Sort

These are the core lab techniques that make biotechnology possible:

- **Restriction Enzyme** — Molecular scissors that cut DNA at a specific sequence; creates fragments that can be rearranged
- **DNA Ligase** — Molecular glue that seals DNA fragments together (you saw this in Module 08 — it also works during DNA replication)
- **Recombinant DNA** — DNA made by combining pieces from two different sources (e.g., a human gene inserted into a bacterial plasmid)

- **Plasmid** — A small, circular piece of DNA found in bacteria; used as a delivery vehicle to carry new genes into cells

PCR — The DNA Photocopier

- **PCR (Polymerase Chain Reaction)** — Makes millions of copies of a specific DNA segment
- It works just like DNA replication (Module 08), but in a test tube:
- **Heat** the DNA to separate the two strands
- **Cool** so short primers attach to the target region
- **Build** new strands using a special heat-proof enzyme (Taq polymerase)
- After each cycle, the number of copies doubles: $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \dots$

Gel Electrophoresis — Sorting DNA by Size

- DNA fragments are placed in a gel and pulled by an electric current
- **Small fragments travel farther**; large fragments stay near the top
- Used to visualize and compare DNA samples

DNA Fingerprinting — Identifying Individuals

- Every person (except identical twins) has a unique pattern of **STRs (Short Tandem Repeats)** — short DNA sequences repeated different numbers of times
- Scientists use PCR to copy STR regions, then gel electrophoresis to compare patterns
- Used in **forensics** (matching crime scene DNA) and **paternity tests** (a child's bands that don't match mom must come from dad)
- This connects to **Module 09 (Inheritance)** — STR patterns are inherited from parents, just like other traits

Genetic Engineering — Moving Genes Between Organisms

- **GMO (Genetically Modified Organism)** — An organism that contains a gene from a different species
- **How it works (simplified):**
- Cut the desired gene out of one organism's DNA (restriction enzyme)

- Paste it into a plasmid (ligase)
- Put the plasmid into a host cell (e.g., bacteria)
- The host cell reads the new gene and makes the protein (central dogma!)
- **Example:** Human insulin gene → inserted into bacteria → bacteria produce insulin for diabetic patients
- **Why this works:** The genetic code is universal (Module 07) — a bacterial ribosome reads human mRNA codons the same way your cells do

CRISPR — Editing the Genome

- **CRISPR-Cas9** — A gene-editing tool that can find a specific DNA sequence and cut, delete, or replace it
- Think of it like **find-and-replace** in a word processor, but for DNA
- **Gene therapy** — Treating genetic diseases by fixing or replacing a broken gene in a patient's cells
- **Ethical questions:** Should we edit genes that get passed to future generations? This is a major ongoing debate

Study Tips

1. **Know the tools by their verb** — Restriction enzymes CUT, ligase GLUES, PCR COPIES, gel SORTS, CRISPR EDITS
2. **PCR = DNA replication in a tube** — It's the same process your cells do (Module 08), just controlled by temperature
3. **Small = far in gel** — Small DNA fragments travel farther through the gel matrix
4. **Universal code = biotech works** — The reason a bacterium can read a human gene is that all life shares the same genetic code (Module 07)
5. **Connect the modules** — This module is the "so what?" for everything since Module 07